

Examples Sheet I

1. Sirius A has an apparent visual magnitude  $m_V = -1.5$  and a spectral type A0V. Its distance is 2.7 pc. Calculate its luminosity and radius in solar units.

Sirius B is a binary companion to Sirius A. It has the same spectral type and  $m_V = 8.5$ . Calculate its luminosity and radius.

Measurements of the orbits of A and B give a binary period of 50 years, mass ratio  $M_A/M_B = 3$  and a semi major-axis of 8 seconds of arc.

Calculate the masses and mean densities of both components. Which star is the main-sequence star and what is the other?

[For an A0V spectrum you may assume  $T_{\text{eff}} = 10,000 \text{ K}$  and Bolometric Correction,  $-(m_{\text{bol}} - m_V) = 0.4$ . An absolute bolometric magnitude  $M_{\text{bol}}$  of zero corresponds to a luminosity of  $3.0 \times 10^{35} \text{ erg s}^{-1}$ . For a binary orbit you may use the fact that the semimajor axis  $a$ , total mass  $M$  and period  $P$  are related by Kepler's law,

$$\left(\frac{a}{\text{au}}\right)^3 = \frac{M}{M_{\odot}} \left(\frac{P}{\text{years}}\right)^2 .]$$

2. In any equilibrium configuration prove that

$$\frac{d}{dr} \left( P + \frac{Gm^2}{8\pi r^4} \right) < 0,$$

where  $m(r) = \int_0^r 4\pi r^2 \rho dr$ . Deduce a lower limit for the central pressure  $P_c$  in terms of the stellar mass and radius. Evaluate the limit for the Sun.

3. Show that the gravitational energy of a spherically symmetric star is

$$\Omega = -4\pi \int_0^R \rho \frac{Gm}{r} r^2 dr.$$

Hence show that a polytrope of index  $n$  has

$$\Omega = -\frac{3}{5-n} \frac{GM^2}{R}.$$

The recombination energy of stellar matter is  $I$  per unit mass. Show that if the internal structure of a star corresponds to an  $n = \frac{3}{2}$  polytrope total disruption of the star is energetically feasible if

$$R > \frac{3}{7} \frac{GM}{I}.$$

Given that the ionization energy of a hydrogen atom is 13.6 eV estimate the maximum radius for a star of  $1 M_{\odot}$  and explain the existence of red giants.

4. By using the virial theorem, or otherwise, show that the period  $\Pi$  of spherically symmetric pulsations of a star satisfying  $I = 4\pi \int_0^R r^4 \rho dr = kMR^2$  and  $\Omega = -\eta GM^2/R$ , where  $k$  and  $\eta$  are constants, is given by

$$\Pi = \left[ \frac{3\pi k}{(3\gamma - 4)\eta G \bar{\rho}} \right]^{\frac{1}{2}},$$

where  $\bar{\rho}$  is the mean density of the star.

Comment on the relation of  $\Pi$  to the free-fall time and the acoustic travel time from surface to centre.

5. If  $\rho$  decreases outwards, show that  $m < \frac{4}{3}\pi r^3 \rho_c$ .

Prove that  $P_c < \frac{1}{2}G \left(\frac{4}{3}\pi\right)^{\frac{1}{3}} \rho_c^{\frac{4}{3}} M^{\frac{2}{3}}$ .

Deduce that if  $\beta = P_{\text{gas}}/P$ , then

$$1 - \beta_c \leq 1 - \beta^*,$$

where  $\beta^*$  satisfies Eddington's quartic,

$$M = \left(\frac{6}{\pi}\right)^{\frac{1}{2}} \left[ \left(\frac{R}{\mu_c}\right)^4 \frac{3}{a} \left(\frac{1 - \beta^*}{\beta^{*4}}\right) \right]^{\frac{1}{2}} G^{-\frac{3}{2}}.$$

Evaluate  $M$  for  $1 - \beta^* = 0.2, 0.5, 0.8$  and comment on what you find.